2.5 Challenges with contract testing

While contract testing offers numerous benefits to software development teams, it's important to acknowledge that it also presents challenges. Understanding and addressing these challenges is crucial for a successful implementation and for reaping the full advantages of contract testing. In this section, we'll discuss some of the common challenges associated with contract testing. Contract testing

- takes time.
- requires a mindset shift.
- requires a lot of buy-in.
- is more technical than other testing activities.
- is difficult to do when there are external teams involved.

2.5.1 Contract testing takes time

From our experience working as software testers, implementing contract testing takes longer than integration or E2E testing. There is a higher bar for adoption to put contract testing into place because of its complexity in the initial setup, which involves setting up the necessary build pipelines, developing contract tests from both the consumer and provider, and integrating them into the development and testing processes. It's crucial to factor in this investment and plan accordingly to ensure a smooth transition.

If we compare the time and effort required for contract tests and E2E tests, as shown in figure 2.10, you can see that while contract tests require more effort, you might not see the benefits immediately. However, contract tests are easier to maintain once this is in place, and the effort over time will decrease. In contrast, E2E tests might be easier to set up initially, but as time progresses and the number of E2E tests increase, it becomes increasingly difficult to maintain and therefore requires a high level of effort the longer you rely on these tests.